SHAW CAMPBELL

Website | campbellshaw8@gmail.com | LinkedIn | GitHub

> Graduate software developer with a BSc Eng (Hons) in Electrical & Computer Engineering from UCT. Experienced across the full stack — from embedded systems and PCB design through to enterprise web applications and deep learning research. Known for picking up new technologies quickly, thriving in Agile teams, and a consistent record of academic distinction, including two Dean's Merit List awards and a merit-based full bursary.

WORK EXPERIENCE

Graduate Software Developer — PSG Financial Services

2026 – Present

Building and maintaining production web applications, working across a .NET / Angular stack with a focus on clean architecture and Agile delivery.

C# · .NET · ASP.NET · Angular · Agile · Web Development

Software Engineering Vacation Work — Peralex Electronics

2025 · 5 weeks

Designed and built a Python desktop application for visualising and navigating openEMS electromagnetic simulation results across multiple antenna types. Participated in the full Agile workflow — sprint planning, stand-ups, and peer code reviews.

Python · OOP · Desktop App Development · Agile

Software Development Intern — Electrum Payments

2024 · 4 weeks

Delivered a real feature on a live company codebase, embedded within the engineering team across stand-ups, planning sessions, and code reviews. Received a full-time offer at the conclusion of the internship.

Java · Spring Boot · OOP · REST APIs · Agile

Signals and Systems II Tutor (EEE3092F)

Jan – Jul 2025 · 4 months

UCT-employed tutor for the third-year signals engineering course, covering Fourier and Laplace transforms, filter design, and discrete-time systems.

Signals and Systems I Tutor (EEE2047F)

Aug – Nov 2025 · 4 months

UCT-employed tutor for the second-year signals engineering course.

Control Systems Laboratory Assistant

2023 & 2024 · 6 weeks per year

Developed, debugged, and maintained software and hardware for a proportional gain control circuit used in undergraduate labs, applying C# and PLC programming alongside hands-on electronics.

Computer Science Tutor — Python (CSC1015F)

2023 · 6 months

UCT-employed tutor for the introductory programming course.

Computer Science Tutor — Java (CSC1016S)

2023 · 6 months

UCT-employed tutor for the introduction to OOP and Java course.

SELECTED PROJECTS

Deep Learning Deformable Registration of Ultrasound to MRI for Prostate Imaging

Thesis · 2025 GitHub

Final-year honours thesis investigating weakly supervised deep learning for deformable registration of transrectal ultrasound (TRUS) and MRI in prostate imaging. Built on a modified VoxelMorph architecture, the work compared intensity-based metrics (MSE, NCC) against label-driven loss functions across 2D and 3D models. Found that unsupervised intensity metrics fail across modalities, while weakly supervised label-similarity models produce anatomically consistent deformation fields. Demonstrated that affine preprocessing via ANTs is essential for stable convergence, and that non-rigid augmentation improves robustness without degrading generalisation.

Python · PyTorch · VoxelMorph · ANTs · Jupyter Notebooks · Medical Image Analysis

Micro-Mouse — Sensing & Power Subsystems

Embedded · 2024 GitHub

Designed and implemented the infrared sensor subsystem and power subsystem for an autonomous maze-solving micro-mouse, interfacing with an existing motherboard and dual-motor drive. Covered the full embedded development loop: KiCad PCB layout for the custom IR sensor board, STM32L476 firmware, power regulation circuitry, and subsystem integration testing.

C · STM32L476 · KiCad · Embedded Systems · PCB Design · Electronics

EDUCATION

BSc Eng (Hons) — Electrical & Computer Engineering

University of Cape Town · 2021 – 2025

- Graduated with Honours (GPA > 70%), thesis in medical machine learning Honours
- *Dean's Merit List* — 2022 & 2023 ×2
- UCT Plus Gold Award — 60 hours of voluntary community service for UCT (2023)
- PSG Full Merit Bursary — 2024 & 2025 Merit

EXTRA-CURRICULARS

Student Residence Mentor

Jan 2022 – Nov 2023 & Jan 2024 – Nov 2024

- Supported first-year students through the academic and social transition into university life across two separate years.

IT & Maintenance Representative — Leo Marquard Hall House Committee

Jan 2023 – Nov 2023

- Built and maintained a website for the campus residence.
- Provided IT support across the residence community.

TECHNOLOGIES

Python | Java | C# | C / C++ | .NET & ASP.NET | Angular | HTML · CSS · JavaScript | Spring Boot | STM32 / Embedded C | PyTorch | Linux | KiCad | Jupyter Notebooks